



IoT

IOTFLOW · n8n

COURSE SLIDES · WSQ

Internet of Things (IoT) Fundamental for Beginners

WSQ Course Code: TGS-2020504020

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

IoT platform: <https://iot.tertiaryinfotech.com>

Version v14 · 6 July 2026

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COURSE ADMINISTRATION

Welcome & Housekeeping

Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer/administrator displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

EXPERTISE

IoT, data analytics, AI/ML and workflow automation (n8n).

DELIVERS

WSQ courses on IoT, AI, data science and software engineering.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

Let's Know Each Other

- Your name and organisation / role.
- Your experience with IoT, electronics or automation (if any).
- What you want to sense, control or automate after this course.

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

5

Be punctual; return from breaks on time.

6

75% attendance is required.

LMS / TMS

- Access your course materials, attendance and assessment on the LMS/TMS portal.
- Portal: <https://ai-lms-tms.tertiaryinfo.tech/>
- Download the slides and Learner Guide for reference during the open-book assessment.

SCHEDULE

Lesson Plan — 2 Days, 8 hours/day

Day 1

- Day 1 — IoT Overview & Collecting Data to the Cloud
 - Digital Attendance (AM) · Introductions
 - Topic 1: Overview of Internet of Things (IoT)
 - Topic 2: Collect and Post Data to Cloud
 - Labs 1–2 on the IoTFlow platform

Day 2 & timing

- Day 2 — Remote Control, Analytics, n8n Automation & Assessment
 - Topic 3: Read Data and Remote Control from Cloud (Labs 3–4)
 - Topic 4: IoT Data Analytics and Visualization (Labs 5–6)
 - Course Feedback and TRAQOM Survey
 - Final Assessment (WA + PP)
- Daily timing
 - 9:30am–6:30pm · 1-hour lunch · tea breaks within

Skills Framework (TSC)

1

TSC Title

Internet of Things Application

2

TSC Code

PTP-TEM-3002-1.1

3

Abilities

A1 briefings on IoT uses & functions · A2 review IoT testing results · A3 integrate multiple data sources · A4 review data for business value.

4

Knowledge

K1 IoT concept · K2 IoT devices · K3 circuits & sensors · K4 wireless technologies · K5 data analytics · K6 cybersecurity.

Learning Outcomes

1

LO1

Understand the uses and functions of IoT technologies.

2

LO2

Post sensor data to cloud for IoT review.

3

LO3

Control devices from cloud data sources.

4

LO4

Data analytics and visualization on cloud to gain business insight.

Course Outline — Day 1

Morning

- Topic 1 Overview of Internet of Things (IoT)
 - (A1, K1, K4)
 - What is IoT?
 - IoT Devices, Sensors and Actuators
 - Triggers and Events
 - Wireless Communication Technologies for IoT
 - IoT Applications and Use Cases

Afternoon

- Topic 2 Collect and Post Data to Cloud
 - (A2, K3)
 - What is Cloud Computing
 - IoT Cloud Platforms — our platform IoTFlow
 - Register a Device and Get a Token
 - Collect Data with Sensors
 - Post Data to Cloud using MQTT or REST API

Course Outline — Day 2

Morning

- Topic 3 Read Data and Remote Control from Cloud
 - (A3, K2)
 - Read Data using MQTT or REST API
 - Remote Control Devices from Cloud
 - Virtual Pins — Blynk-style two-way control
 - Alerts and Trigger Rules

Afternoon

- Topic 4 IoT Data Analytics and Visualization
 - (A4, K5, K6)
 - Visualize IoT Data on Cloud Dashboards
 - Analyze IoT Data for Business Insight
 - Automate with n8n Workflows
 - AI for IoT Analytics
 - Cyber Security Concern

Criteria for Funding

- Minimum attendance rate of 75% based on the SSG Digital Attendance record.
- Complete the assessment and be assessed as 'Competent'.

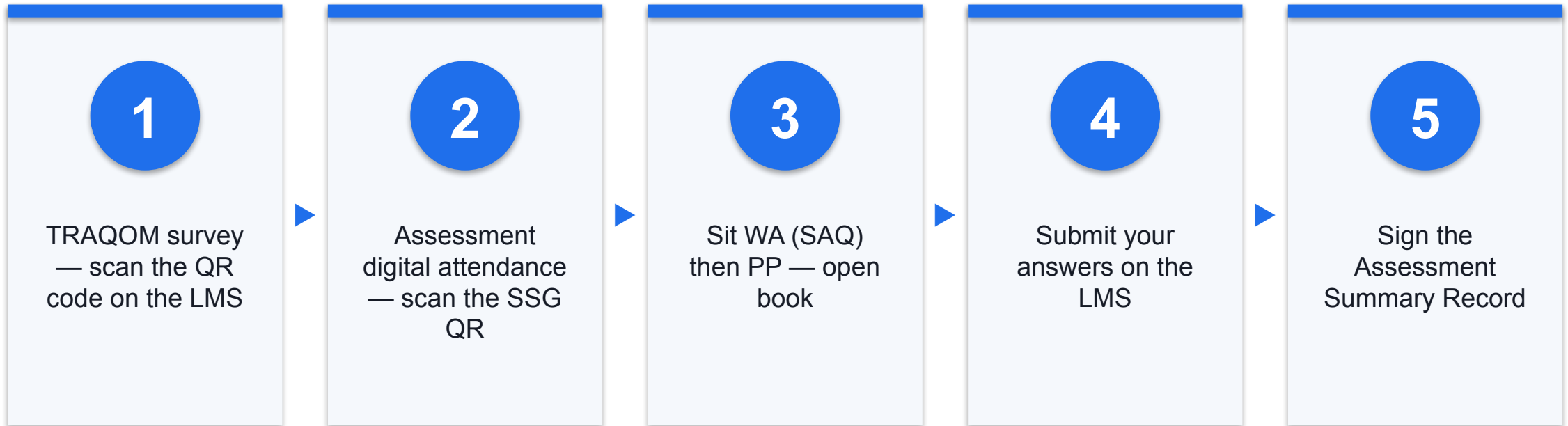
Briefing for Assessment

- Place phones and other materials under the table or on the floor.
- No photos or recording of assessment scripts.
- No discussion during the assessment.
- Use a black/blue pen for hard-copy assessments.
- No liquid paper / correction tape.
- Scripts are collected when time is up.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — hands-on IoT platform tasks, 1 hour, open book.
- Format: Open Book — slides, Learner Guide and approved materials only.
- A minimum of 75% attendance is required to be eligible for assessment and funding.
- An appeal process is available if required.

Assessment Flow





OUR IOT PLATFORM

IoTFlow — iot.tertiaryinfotech.com

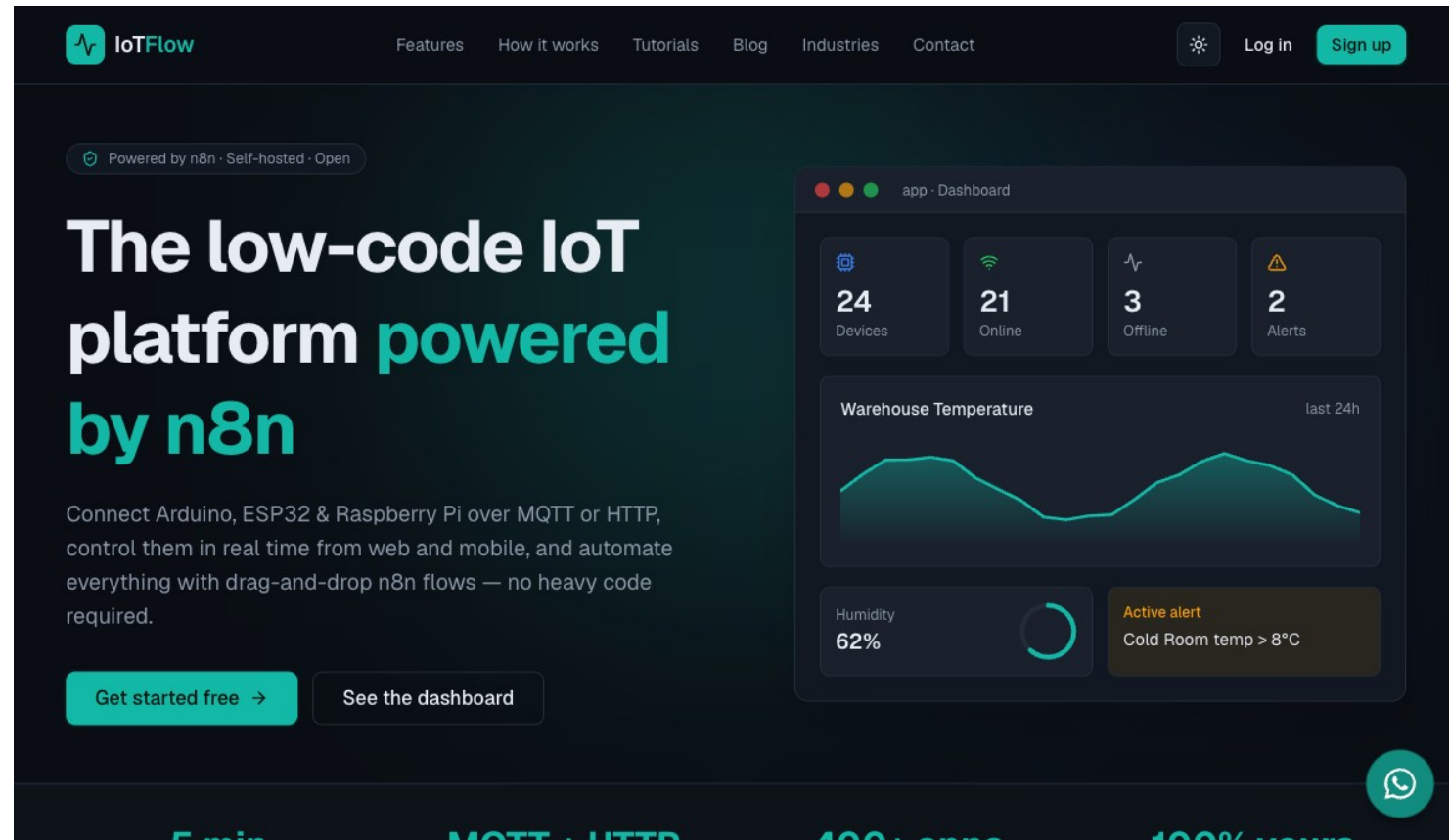
The low-code IoT platform you will use for every lab in this course

IoTFlow · <https://iot.tertiaryinfotech.com>

Connect. Visualise. Automate.

Connect Arduino, ESP32 & Raspberry Pi over MQTT or HTTP, control them in real time from web and mobile, and automate everything with drag-and-drop n8n flows.

IoTFlow — Low-code IoT Platform powered by n8n



Sign up free at <https://iot.tertiaryinfotech.com> — you will use it in every lab.

How it Works

1

Connect — flash the wizard snippet to your ESP32 / Arduino / Pi. It streams data over MQTT or HTTP.

2

Visualise & Control — compose a dashboard of charts, gauges, buttons & sliders, on web and mobile.

3

Automate with n8n — device events fire n8n flows that notify, control devices back, log data or call AI.

IoTFlow Features (1 of 2)

1

Device Connectivity

Guided wizard with copy-paste code for ESP32, Arduino, Raspberry Pi, C++ & Python over MQTT or HTTP.

2

Two-way Control

Buttons, switches & sliders write to virtual pins — control relays, motors & LEDs live, Blynk-style.

3

Low-code Automation

Drag-and-drop n8n flows react to device events — notify, control, log, or call AI with zero code.

4

Dashboard

Live telemetry with auto-refresh, number cards, charts, gauges, LED indicators & maps.

IoTFlow Features (2 of 2)

1

Messaging Protocols

A managed MQTT broker and a simple REST endpoint. Uplink telemetry and downlink commands.

2

Alerts

Trigger on thresholds or device-offline, then hand off to an n8n flow.

3

Cross-platform Apps

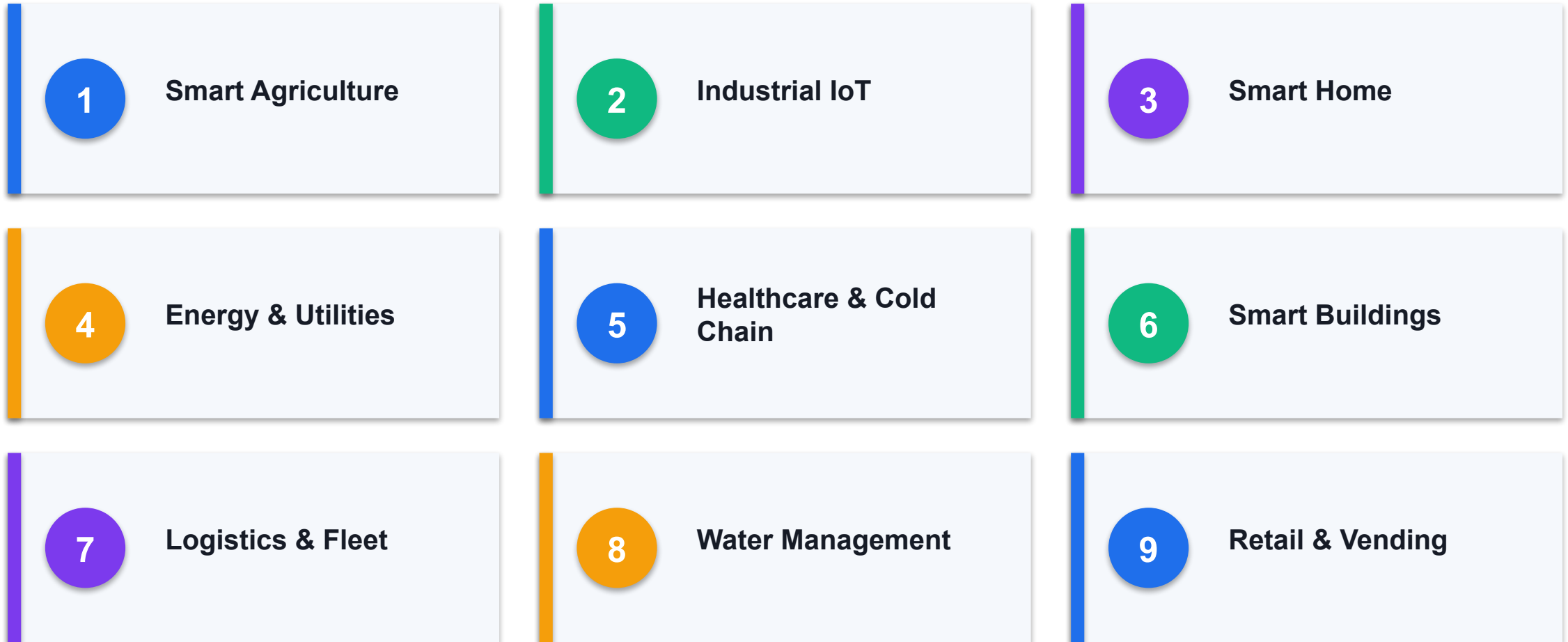
Installable PWA — add IoTFlow to your phone's home screen for mobile control.

4

Self-hosted

Docker & Coolify-ready. Own your data and your automation engine.

Industries Served



Platform Tutorials → Course Labs

Get connected

- Tutorial: Add your first device → Lab 1
- Tutorial: Send your first reading → Lab 2

Read & control

- Read state via REST & MQTT → Lab 3
- Tutorial: Control a device → Lab 4

Visualise & automate

- Tutorial: Build a dashboard → Lab 5
- Tutorial: Automate with n8n → Lab 6

TOPIC 01

Overview of Internet of Things (IoT)

What is IoT · Devices, Sensors & Actuators · Triggers · Wireless Technologies · Use Cases

Key Concepts — Overview of Internet of Things (IoT)

1

IoT is the network of physical devices embedded with sensors, software and connectivity that collect and exchange data over the internet.

2

A device senses the world with sensors (input) and acts on it with actuators (output) — a trigger turns a reading into an event that starts an action.

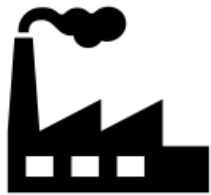
3

Wireless technologies (Wi-Fi, Bluetooth, Zigbee, LoRaWAN, NB-IoT, 5G) connect constrained devices to the cloud.

4

IoT powers smart homes, agriculture, healthcare, manufacturing, logistics, retail and smart cities.

The 4th Industrial Revolution



Industry 1.0

Mechanization,
steam power,
weaving loom

(18th century,
1760 - 1840)



Industry 2.0

Mass production,
assembly line,
electrical energy

(19th century,
1871 - 1914)



Industry 3.0

Automation,
computers and
electronics

(late 20th century,
1970 - present)



Industry 4.0

Cyber Physical
Systems, internet of
things, networks

(early 21st century,
2011 - present)

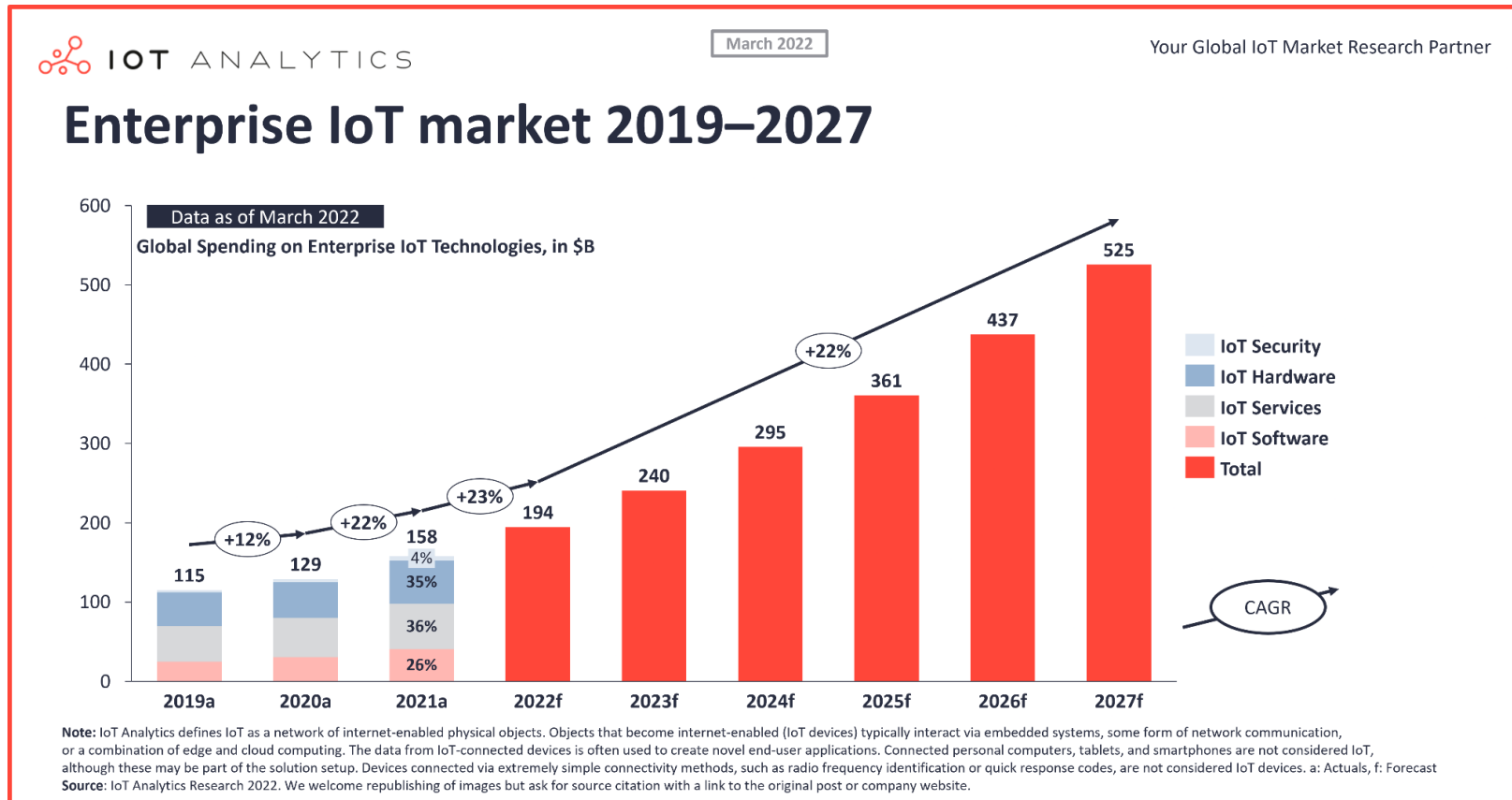
Industry 4.0: cyber-physical systems — IoT, big data, cloud, AI and automation.

DEFINITION

What is the Internet of Things (IoT)?

- IoT refers to the network of physical devices, vehicles, buildings and other objects embedded with sensors, software and connectivity.
- These 'things' collect and exchange data over the internet — without any human interaction.
- Low-cost sensors and long-range wireless technology are instrumental to IoT; 5G drives mass adoption.
- By connecting the physical world to the cloud, IoT turns real-world events into data you can analyse and act on.

Internet of Things — Everything Connected



Devices, people and platforms exchanging data over the internet.

IoT Components



Sensors

Collecting data



Connectivity

Sending data to cloud



Data Processing

Making data useful

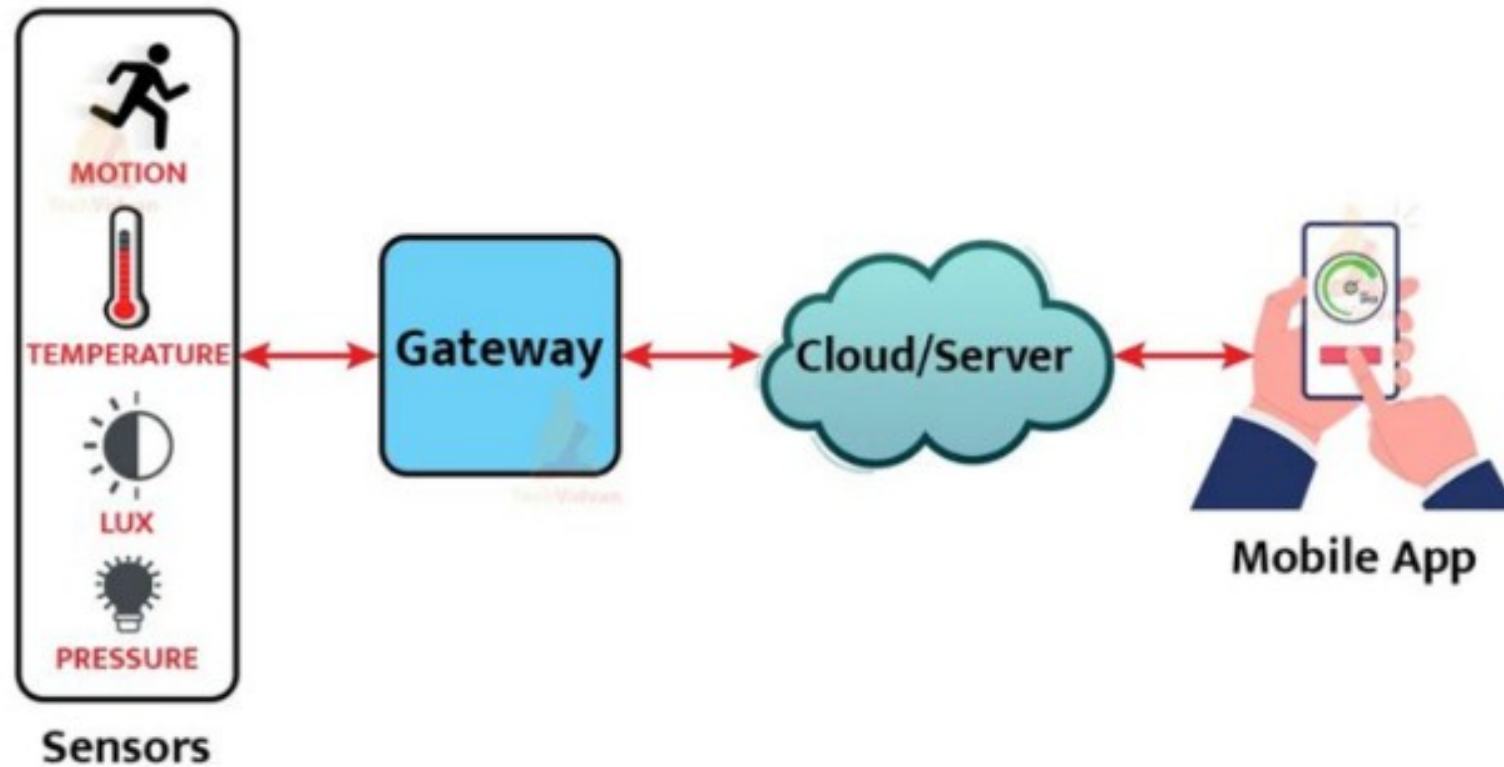


User Interface

Delivering information to user

A 'thing' = embedded electronics + sensors + connectivity + data.

How Does IoT Work?



Sensing devices stream data to an IoT platform, which integrates, analyses and shares the most valuable information.

What Are IoT Devices?

1

Physical objects that sense

Hardware such as sensors, gadgets, appliances and machines that collect and exchange data over the internet.

2

Programmed for a purpose

Each device is built for a specific application and can be embedded into other IoT devices.

3

Connected & addressable

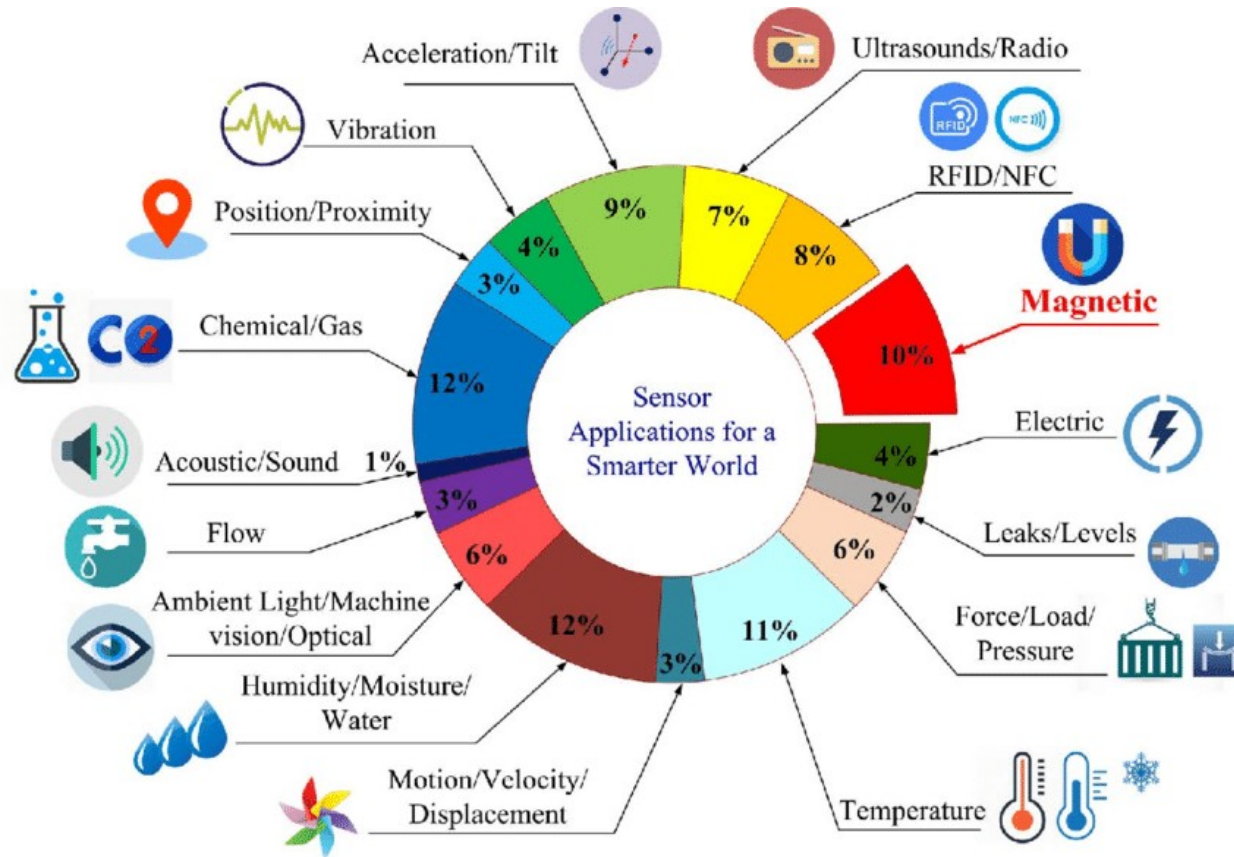
Integrated CPU, network adapter and firmware, with an IP address on the network.

4

Managed by software

Configured and controlled through an app or dashboard — e.g. your phone controls the lights at home.

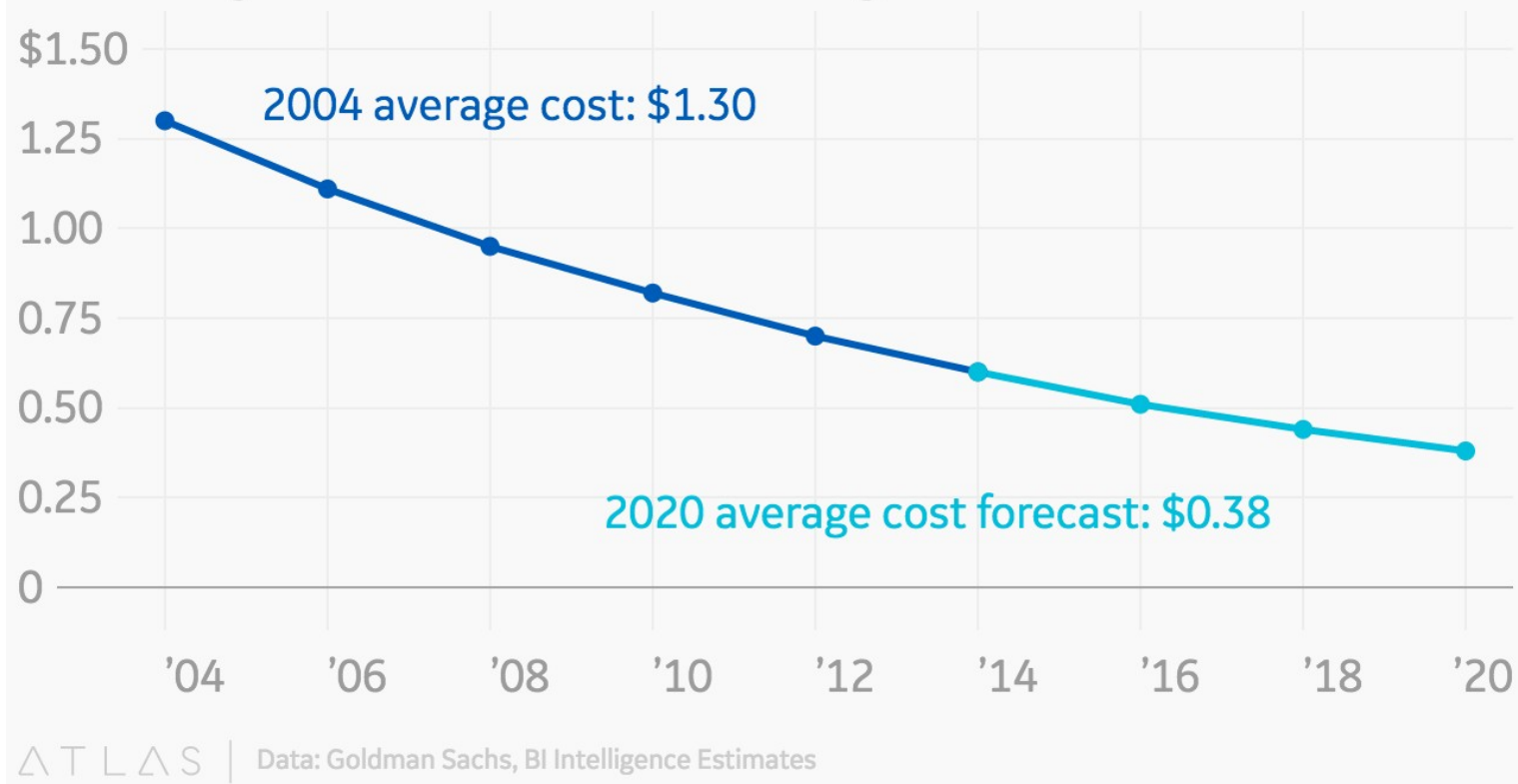
Sensors for IoT



Temperature, humidity, IR, proximity, gas, chemical, motion, light — sensors are the device's senses.

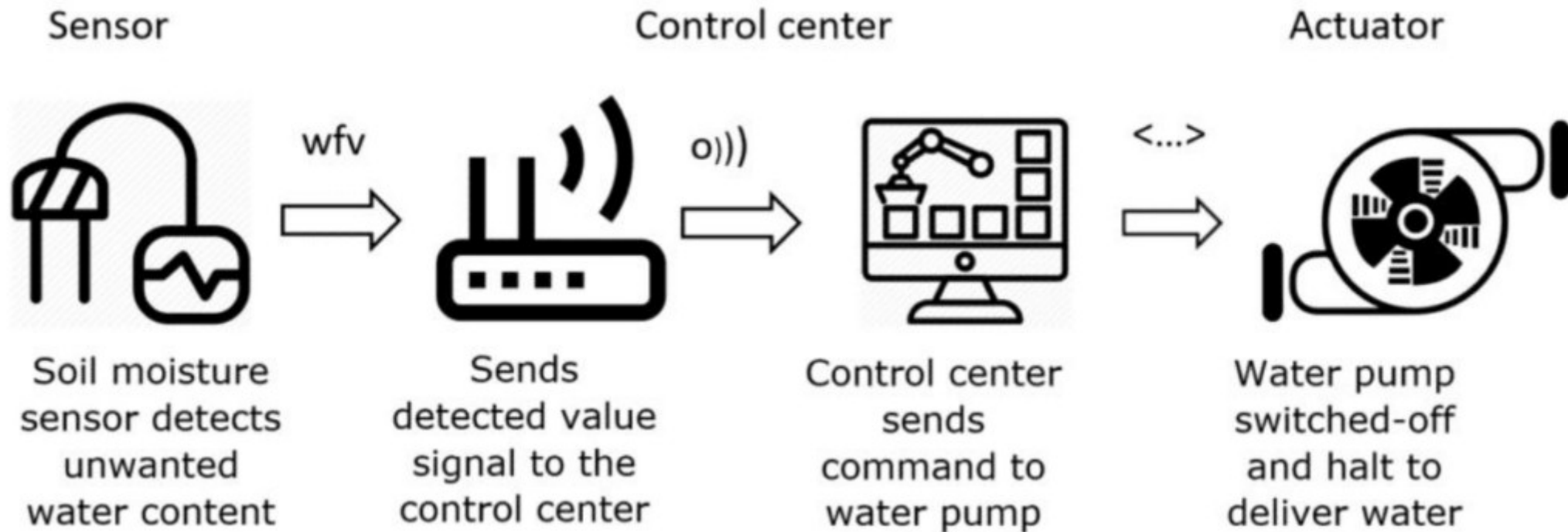
More Sensors for IoT

The average cost of IoT sensors is falling



Pick the sensor for the physical quantity you need to measure.

IoT Actuators



Actuators act on the world: relays, motors, pumps, valves, LEDs — the output side of IoT.

Sensors → Triggers → Actions

1

Sense

A sensor produces a reading — temperature 31 °C, door open, tank level low.

2

Trigger

A rule watches the reading — 'temperature above 30' or 'device offline' fires an event.

3

Act

The event drives an action — notify someone, log it, call AI, or command an actuator.

4

Automate

On IoTFlow, triggers hand off to n8n workflows — the full loop with zero code (Topic 4).

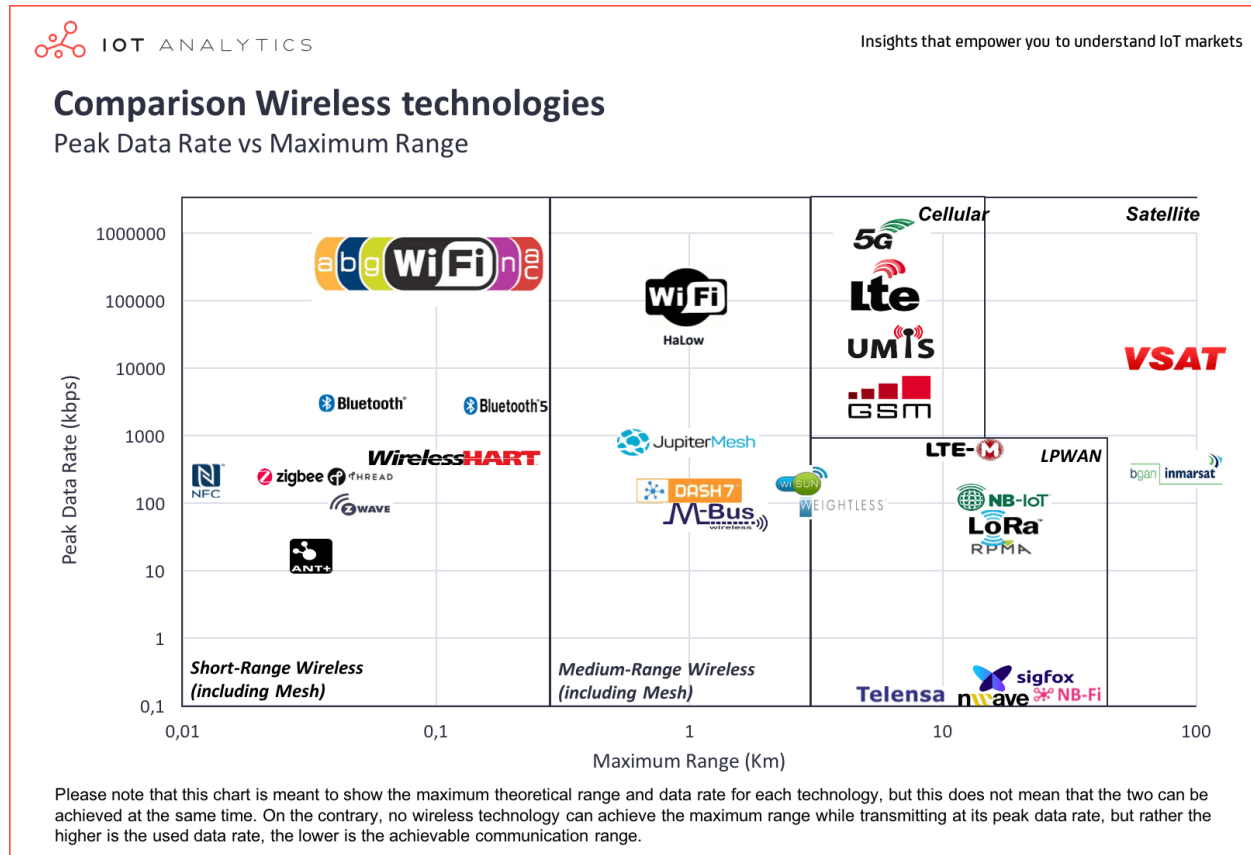
Wireless Technologies for IoT (Short Range)

- Wi-Fi — widely used, high data rates and reliable connectivity over short-to-medium distances.
- Bluetooth — low-power, short-range; connects IoT devices to phones and tablets.
- Zigbee — low-power wireless mesh for low data rates and reliable longer-distance links.
- Z-Wave — optimised for low-power devices; reliable communication over longer distances.

Wireless Technologies for IoT (Long Range)

- LoRaWAN — long-range, low-power; designed for smart-city-scale applications.
- Sigfox — LPWAN for low data rates and long range, e.g. asset tracking and remote monitoring.
- NB-IoT — narrowband cellular LPWAN with reliable long-range connectivity.
- 5G — high bandwidth and massive device density; the driver for IoT mass adoption.

Comparison of Wireless Technologies



Trade-off: range vs bandwidth vs power consumption.

Types of IoT

1

Consumer IoT

Everyday use — home appliances, voice assistants, light fixtures.

2

Commercial IoT

Healthcare and transport — smart pacemakers, monitoring, connected cars.

3

Industrial IoT (IIoT)

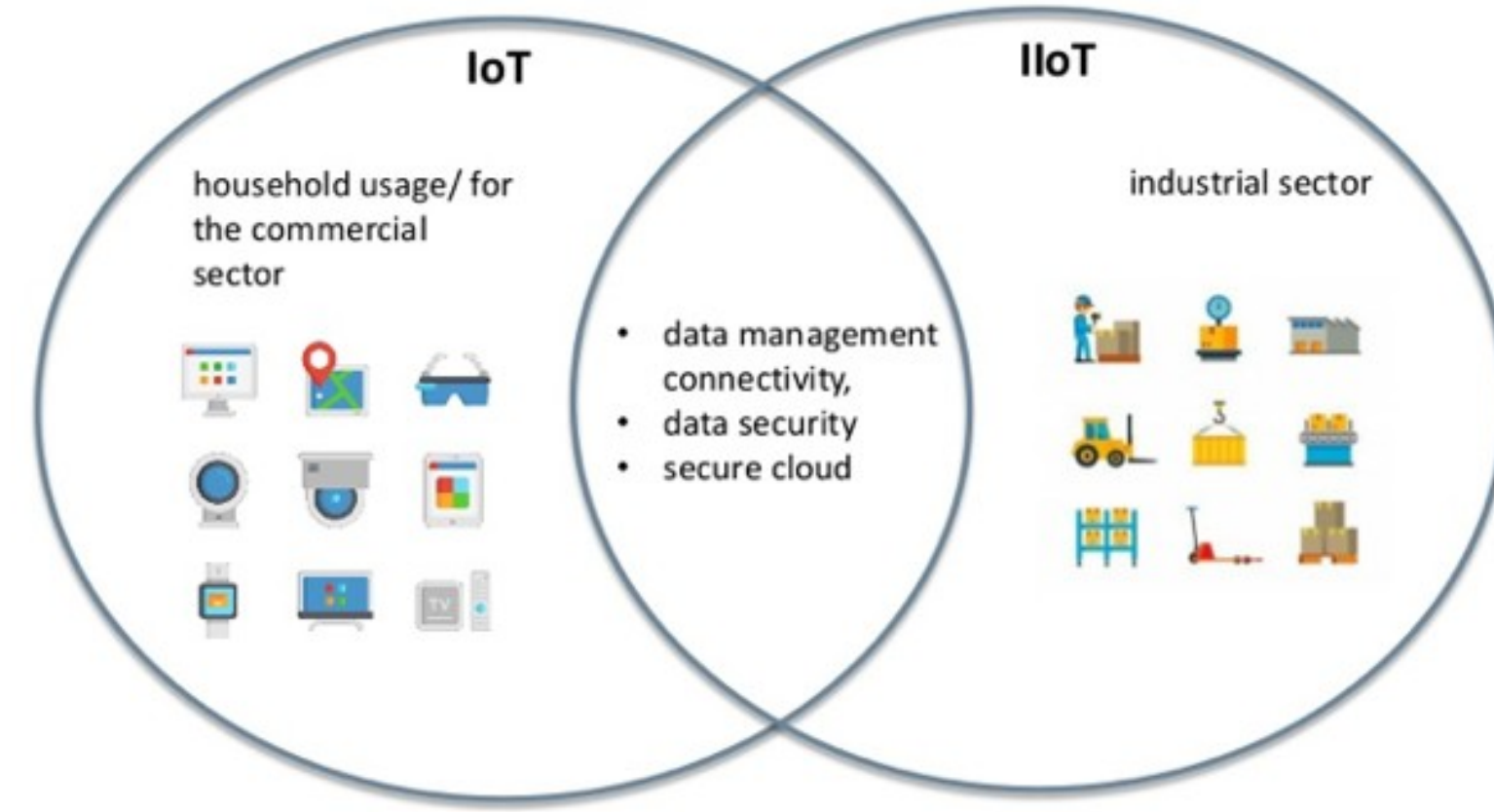
Manufacturing and energy — digital control systems, smart agriculture, industrial big data.

4

Infrastructure IoT

Smart-city connectivity — infrastructure sensors and management systems.

IoT vs Industrial IoT (IIoT)



IIoT extends IoT into industrial sectors — real-time data from sensors to equipment improves business processes.

IoT Applications



IoT has found its way into every conceivable field — from agriculture to aerospace.

Use Case — Healthcare



Track patient-assistance assets: IoT-tagged wheelchairs located in seconds.

USE CASES

Use Case — Smart Home



Smart outlets, thermostats, cameras and locks — efficiency, safety and comfort.

Use Case — Agriculture



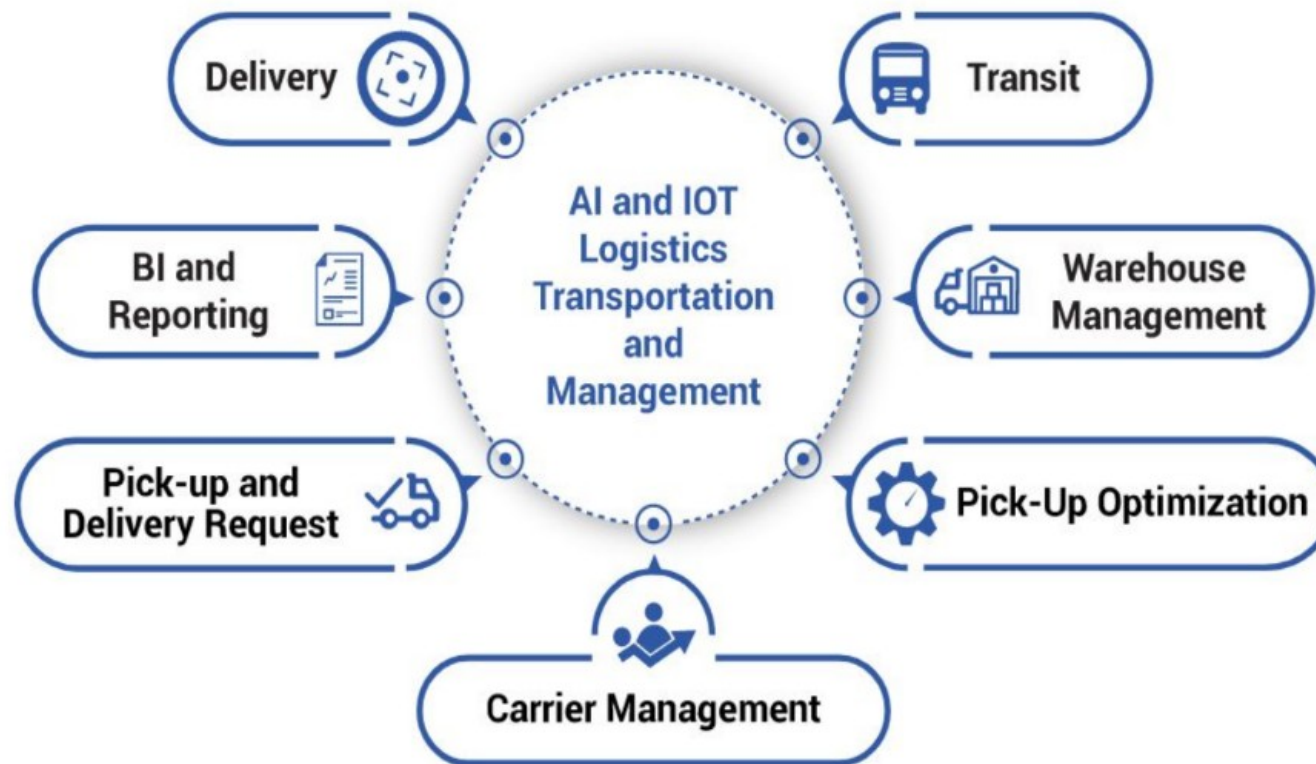
Soil moisture, weather and crop sensors drive precision irrigation.

Use Case — Manufacturing



Production-line monitoring enables proactive maintenance before failure.

Use Case — Logistics & Smart City



Fleets rerouted on live conditions; cities manage traffic, utilities and environment.

Recap — Overview of Internet of Things (IoT)

- IoT connects physical devices with sensors, software and connectivity to collect and exchange data.
- Devices sense with sensors, act with actuators, and triggers turn readings into events and actions.
- Short-range (Wi-Fi, BLE, Zigbee) and long-range (LoRaWAN, NB-IoT, 5G) wireless link devices to the cloud.
- IoT use cases span smart home, healthcare, agriculture, manufacturing, logistics and smart cities.

Lunch Break

1 hour

TOPIC 02

Collect and Post Data to Cloud

Cloud & IoT Platforms · IoTFlow · Device Registration · MQTT & HTTP Telemetry

Key Concepts — Collect and Post Data to Cloud

1

The cloud stores and processes IoT data at scale — devices stream telemetry to an IoT platform instead of a single computer.

2

IoTFlow (iot.tertiaryinfotech.com) is our low-code IoT platform: connect ESP32/Arduino/Pi over MQTT or HTTP, visualise, control and automate.

3

Every device authenticates with a device token (Authorization: Bearer dev_...) issued once at registration.

4

Telemetry is a simple JSON document of metric key-value pairs, e.g. {"temperature": 22.5, "humidity": 60}.

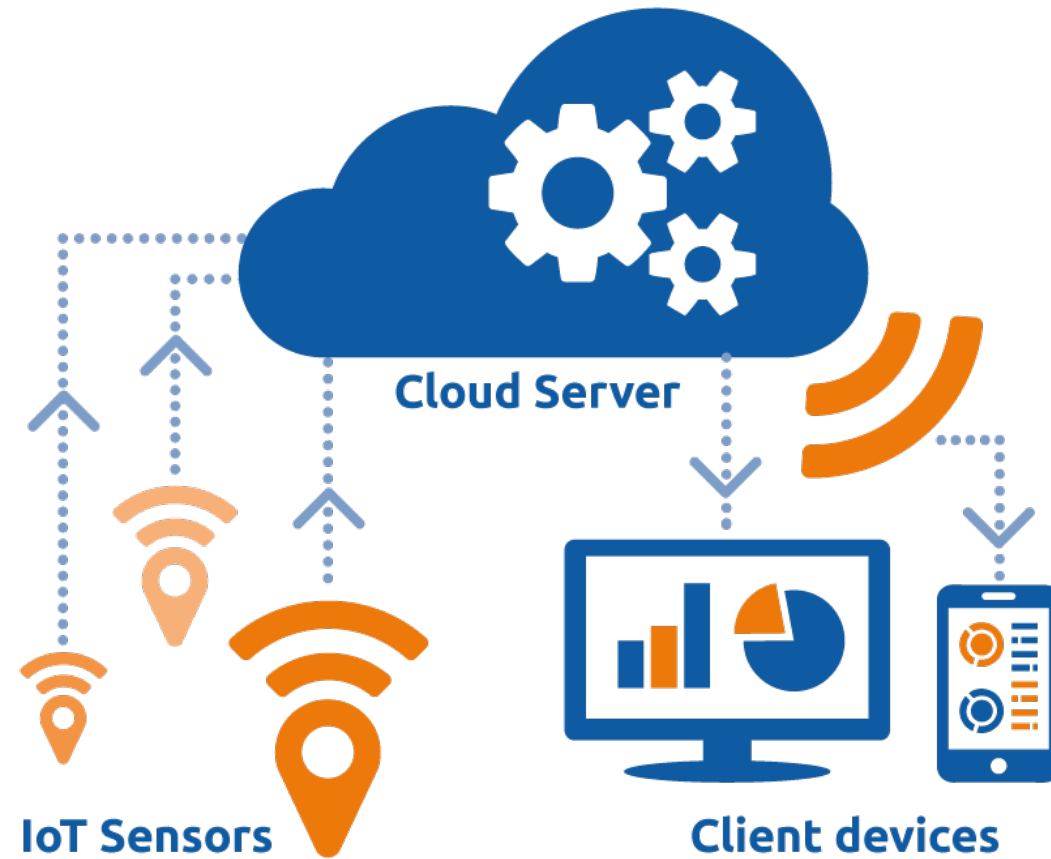
5

Official device clients: the Python client — `pip install iotflow` — for Raspberry Pi/PC/Mac, and the Arduino library for ESP8266/ESP32; both speak the same protocol.

6

The Python client sends with `send()/virtual_write()` over HTTP, streams with `connect()/mqtt_publish()` over MQTT, and reacts to commands with `@on_command`.

Cloud Computing for IoT



The cloud stores, processes and serves IoT data at scale — pay-as-you-go, no servers to run.

IoT Cloud Platforms

1

IoTFlow (this course)

Our low-code platform — MQTT/HTTP, dashboards, virtual pins, n8n automation, AI.

2

ThingSpeak

MATLAB-centric IoT analytics platform with channels and REST/MQTT APIs.

3

AWS IoT / Azure IoT / Google

Hyperscaler IoT suites for enterprise fleets and big-data pipelines.

4

Blynk / Arduino Cloud

Maker-friendly device platforms with mobile apps and virtual pins.

Two Ways to Post Data to the Cloud

- HTTP REST API — the device POSTs a JSON document to an endpoint; simplest to test from any terminal.
- MQTT — a lightweight publish/subscribe protocol over TCP/IP for bandwidth- and power-constrained devices.
- On IoTFlow both are first-class: a managed MQTT broker (port 1883) and a REST endpoint (/api/telemetry).
- Either way, every message is authenticated with your device token: Authorization: Bearer dev_...

MQTT in a Nutshell

1

Publish / Subscribe

Clients publish messages to topics; subscribers receive them — no polling.

2

Broker

The central post office: dispatches every message from senders to the right receivers.

3

Topics

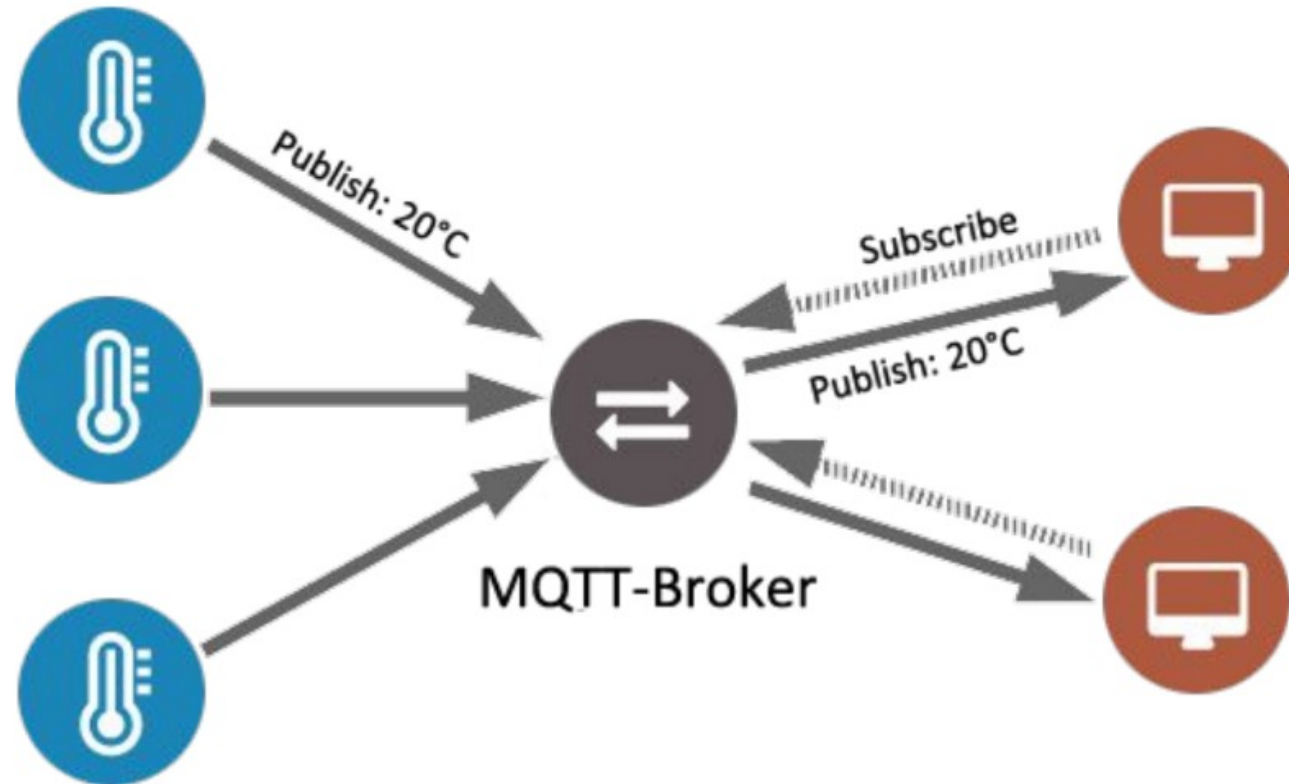
Routing keys such as devices/<id>/up — publish to a topic, subscribe to a topic.

4

Lightweight

Designed for constrained devices and unreliable networks; runs over TCP/IP or WebSockets.

How MQTT Works



Publishers → broker → subscribers, routed by topic.

The IoTFlow Python Client — pip install iotflow

1

Install

pip install "iotflow[mqtt]" for real-time MQTT, or pip install iotflow for HTTP only (zero dependencies). For Raspberry Pi, PC, Mac and Linux SBCs.

2

Report readings (HTTP)

iot = IoTFlow(host, token, device_id) then
iot.send(temperature=22.5, humidity=60) — or one metric at a time with iot.virtual_write().

3

Stream readings (MQTT)

iot.connect() runs MQTT in the background;
iot.mqtt_publish(...) in a loop — the telemetry_upload.py example publishes every 10 s.

4

React to commands

Decorate a handler with @iot.on_command(pin, value, text);
iot.run(interval=3) polls over HTTP, iot.loop_forever() listens over MQTT.

Python Client — Downloads & References

- PyPI package: `pip install iotflow` · <https://pypi.org/project/iotflow/>
- Package folder + README: <https://github.com/alfredang/iotplatform/tree/main/clients/python>
- Example script (the Arduino replacement): `clients/python/examples/telemetry_upload.py`
- Direct download: `curl -O https://raw.githubusercontent.com/alfredang/iotplatform/main/clients/python/examples/telemetry_upload.py`
- The Arduino example it mirrors: `clients/arduino/IoTFlow/examples/ESP8266_Telemetry_Upload`
- Run it: `pip install "iotflow[mqtt]"`, fill in your broker host + device id + token, then: `python telemetry_upload.py`

Hands-On Labs — Topic 02

1

Lab 1 — Register Your First Device on IoTFlow

A registered IoT device in your own IoTFlow project, with its device token saved

2

Lab 2 — Send Sensor Readings to the Cloud (HTTP & MQTT)

Live temperature & humidity telemetry streaming into IoTFlow from terminal, Python and ESP32

Register Your First Device on IoTFlow

LAB 1

The learner signs in to the IoTFlow platform, creates a project workspace, runs the Add Device wizard, chooses a protocol (HTTP or MQTT) and obtains the device token used to authenticate every message the device sends.

You'll build

A registered IoT device in your own IoTFlow project, with its device token saved

Uses: IoTFlow projects, Add Device wizard, device tokens, MQTT & HTTP protocols



Register Your First Device on IoTFlow

STEP 1 OF 6

1

Open the IoTFlow platform and sign up for a free account (or log in).

<https://iot.tertiaryinfotech.com> → Sign up → Log in

Register Your First Device on IoTFlow

STEP 2 OF 6

A green circle with a white number 2 inside, indicating the second step in a sequence.

Pick or create a project — a self-contained workspace with its own dashboard, alerts, map and automations. Use the project switcher at the top of the sidebar.

Register Your First Device on IoTFlow

STEP 3 OF 6

3

Click Add Device in the sidebar and give your device a descriptive name.

Device name: Living Room Sensor

Register Your First Device on IoTFlow

STEP 4 OF 6



4

Follow the six-step guided wizard and choose a protocol: HTTP — simplest, POST JSON to a REST endpoint; or MQTT — lightweight pub/sub, ideal for many devices and real-time control. Both remain available later.

Register Your First Device on IoTFlow

STEP 5 OF 6

5

Copy the device token shown when the device is created — it is displayed only once.

```
dev_XXXXXXXXXXXX
```

Register Your First Device on IoTFlow

STEP 6 OF 6

6

Understand how the token authenticates every message your device sends.

```
Authorization: Bearer dev_XXXXXXXXXXXX
```

Register Your First Device on IoTFlow

Test it

The new device appears in your project's device list, and the Integrate page shows code snippets pre-filled with your endpoint and device token.

Send Sensor Readings to the Cloud (HTTP & MQTT)

LAB 2

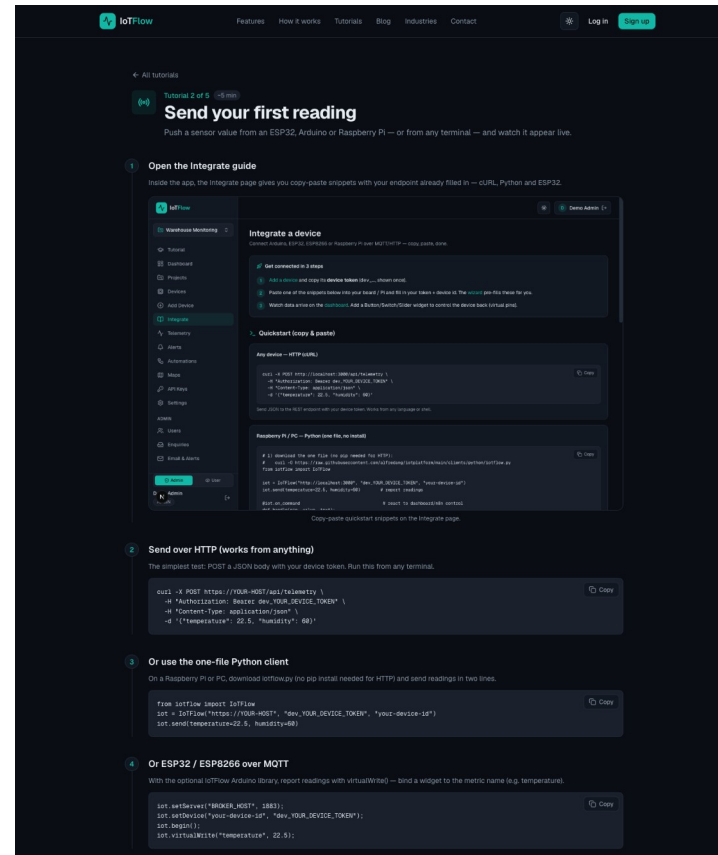
The learner pushes sensor values to the IoTFlow telemetry API three ways — cURL from any terminal, the Python client, and an ESP32/ESP8266 over MQTT — and watches the readings appear live on the dashboard.

You'll build

Live temperature & humidity telemetry streaming into IoTFlow from terminal, Python and ESP32

Uses: Telemetry REST API, Python client (pip install iotflow), ESP32/ESP8266 Arduino library, MQTT broker (port 1883)

Send Sensor Readings to the Cloud (HTTP & MQTT)



Follow along on the platform: <https://iot.tertiaryinfotech.com/tutorials/send-your-first-reading>

Send Sensor Readings to the Cloud (HTTP & MQTT)

STEP 1 OF 8



Open the Integrate page for your device — every snippet comes pre-filled with your endpoint and device token.

Send Sensor Readings to the Cloud (HTTP & MQTT)

STEP 2 OF 8

2

Send your first reading from any terminal with cURL (HTTP POST).

```
curl -X POST https://iot.tertiaryinfotech.com/api/telemetry \  
-H "Authorization: Bearer dev_XXXXXXXXXXXX" \  
-H "Content-Type: application/json" \  
-d '{"temperature": 22.5, "humidity": 60}'
```

Send Sensor Readings to the Cloud (HTTP & MQTT)

STEP 3 OF 8

3

Open the Dashboard — the reading appears under Latest Telemetry within seconds (widgets auto-refresh every 5 s).

Send Sensor Readings to the Cloud (HTTP & MQTT)

STEP 4 OF 8

4

Install the official IoTFlow Python client from PyPI (<https://pypi.org/project/iotflow/>).

```
pip install "iotflow[mqtt]" # with real-time MQTT (paho-mqtt)
pip install iotflow        # HTTP only — zero dependencies
```

Send Sensor Readings to the Cloud (HTTP & MQTT)

STEP 5 OF 8

5

Send readings from Python over HTTP — several metrics at once with `send()`, or one at a time with `virtual_write()`.

```
from iotflow import IoTFlow
```

```
iot = IoTFlow("https://iot.tertiaryinfotech.com", "dev_XXXXXXXXXXXX", "living-room-sensor")
```

```
iot.send(temperature=22.5, humidity=60)    # several metrics at once
```

```
iot.virtual_write("temperature", 22.5)    # or one at a time
```

Send Sensor Readings to the Cloud (HTTP & MQTT)

STEP 6 OF 8

6

Stream continuously over MQTT with the `telemetry_upload.py` example — download it, fill in your broker host, device id and device token, then run it.

```
curl -O https://raw.githubusercontent.com/alfredang/iotplatform/main/clients/python/examples/telemetry_upload.py  
# edit: MQTT_BROKER = "iot.tertiaryinfotech.com", DEVICE_ID, DEVICE_TOKEN  
python telemetry_upload.py
```

Send Sensor Readings to the Cloud (HTTP & MQTT)

STEP 7 OF 8

7

Study the loop inside `telemetry_upload.py` — `connect()` runs MQTT in the background and `mqtt_publish()` sends a reading every 10 seconds.

```
iot = IoTFlow(token=DEVICE_TOKEN, device_id=DEVICE_ID,
               mqtt_host=MQTT_BROKER, mqtt_port=MQTT_PORT)
iot.connect()          # MQTT runs in the background
while True:
    iot.mqtt_publish(temperature=temperature, humidity=humidity, voltage=3.7)
    time.sleep(PUBLISH_INTERVAL_S)
```

Send Sensor Readings to the Cloud (HTTP & MQTT)

STEP 8 OF 8

8

Hardware alternative: flash the ESP8266_Telemetry_Upload Arduino sketch — the Arduino library speaks the same protocol as the Python client. Confirm the values update live on the dashboard.

```
IoTFlow.virtualWrite("temperature", t); // metric name binds to dashboard widgets
```

Send Sensor Readings to the Cloud (HTTP & MQTT)

Test it

Each value you send appears in the Dashboard's Latest Telemetry within seconds, and the device's telemetry history grows with every reading.

Recap — Collect and Post Data to Cloud

- You registered a device on IoTFlow and secured its once-shown device token.
- You posted telemetry with cURL (HTTP) and the Python client — `pip install iotflow` — over HTTP and MQTT.
- The `telemetry_upload.py` example is the Python equivalent of the ESP8266 Arduino sketch; both speak the same protocol.
- Telemetry is JSON metric key-value pairs; metric names bind to dashboard widgets.
- Readings appear in Latest Telemetry within seconds — auto-refresh every 5 s.

TOPIC 03

Read Data and Remote Control from Cloud

Read via REST API & MQTT · Virtual Pins · Dashboard Control · Alerts & Triggers

Key Concepts — Read Data and Remote Control from Cloud

1

Reading from the cloud integrates data sources: REST GET for state and history, MQTT subscribe for real-time push.

2

Virtual pins (V1, relay, pump ...) are named keys that dashboard control widgets write to — buttons, switches, sliders and terminals.

3

Devices receive downlink commands on the MQTT topic `devices/<id>/down`, or by HTTP-polling `GET /api/device/state`.

4

Alert rules fire on metric thresholds or device-offline, and can hand off to an n8n automation.

Reading Data from the Cloud

HTTP GET

- REST API (pull)
 - GET /api/device/state returns latest metrics & pins
 - Simple from any language — curl, Python requests
 - Per-device telemetry history on the platform

MQTT Subscribe

- MQTT (push)
 - Subscribe to your device's topics
 - Messages arrive the moment they are published
 - Ideal for real-time apps and integrations

Virtual Pins — Two-Way Control

1

What they are

Named keys (V1, relay, pump) shared between dashboard widgets and device code — Blynk-style.

2

Control widgets

Button (momentary) · Switch (toggle) · Slider (value) · Terminal (text).

3

Downlink path

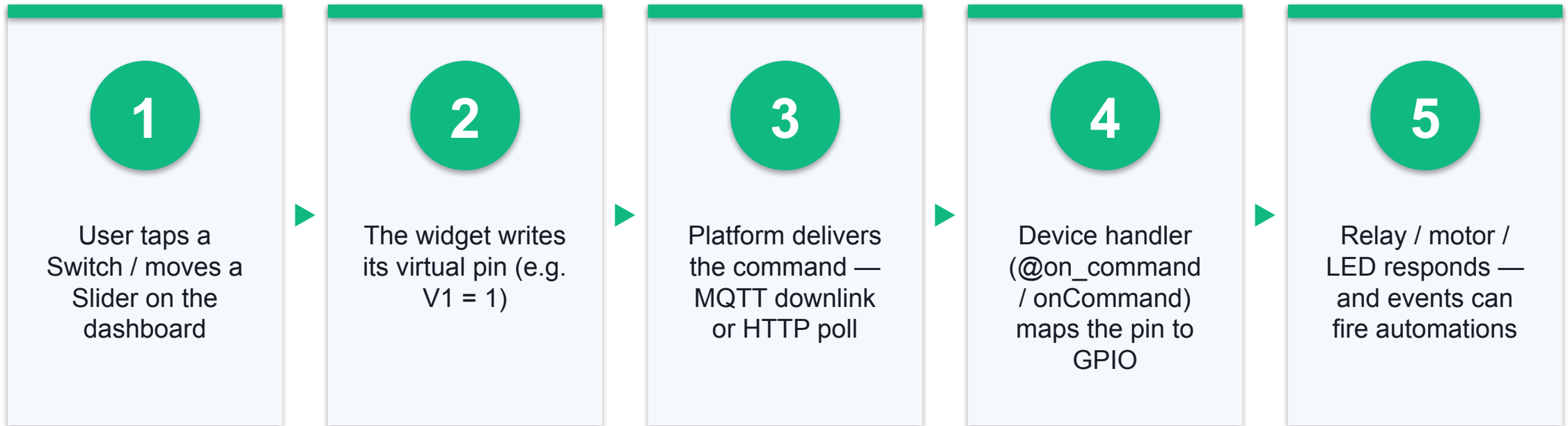
Widget writes a pin → platform sends the command → device receives on `devices/<id>/down` (MQTT) or polls `/api/device/state` (HTTP).

4

Device handler

A handler maps the pin to a GPIO — Python `@iot.on_command(pin, value, text)` or Arduino `onCommand()`; the relay, motor or LED responds instantly.

Remote Control — the Downlink Path



Alerts and Trigger Rules

- Alert rules watch your telemetry: fire on a threshold (temperature > 30) or when a device goes offline.
- Active alerts are tracked per device and can be shown in an Alert-list dashboard widget.
- An alert can hand off to an n8n automation — notify, escalate, log or control a device back (Topic 4).
- Triggers are the bridge from monitoring to automation.

Hands-On Labs — Topic 03

3

Lab 3 — Read Device Data from the Cloud (REST API & MQTT)

A terminal and Python session reading live device state and streaming MQTT messages

4

Lab 4 — Remote Control a Device with Dashboard Virtual Pins

A dashboard switch and slider that control a relay/LED on the device in real time

Read Device Data from the Cloud (REST API & MQTT)

LAB 3

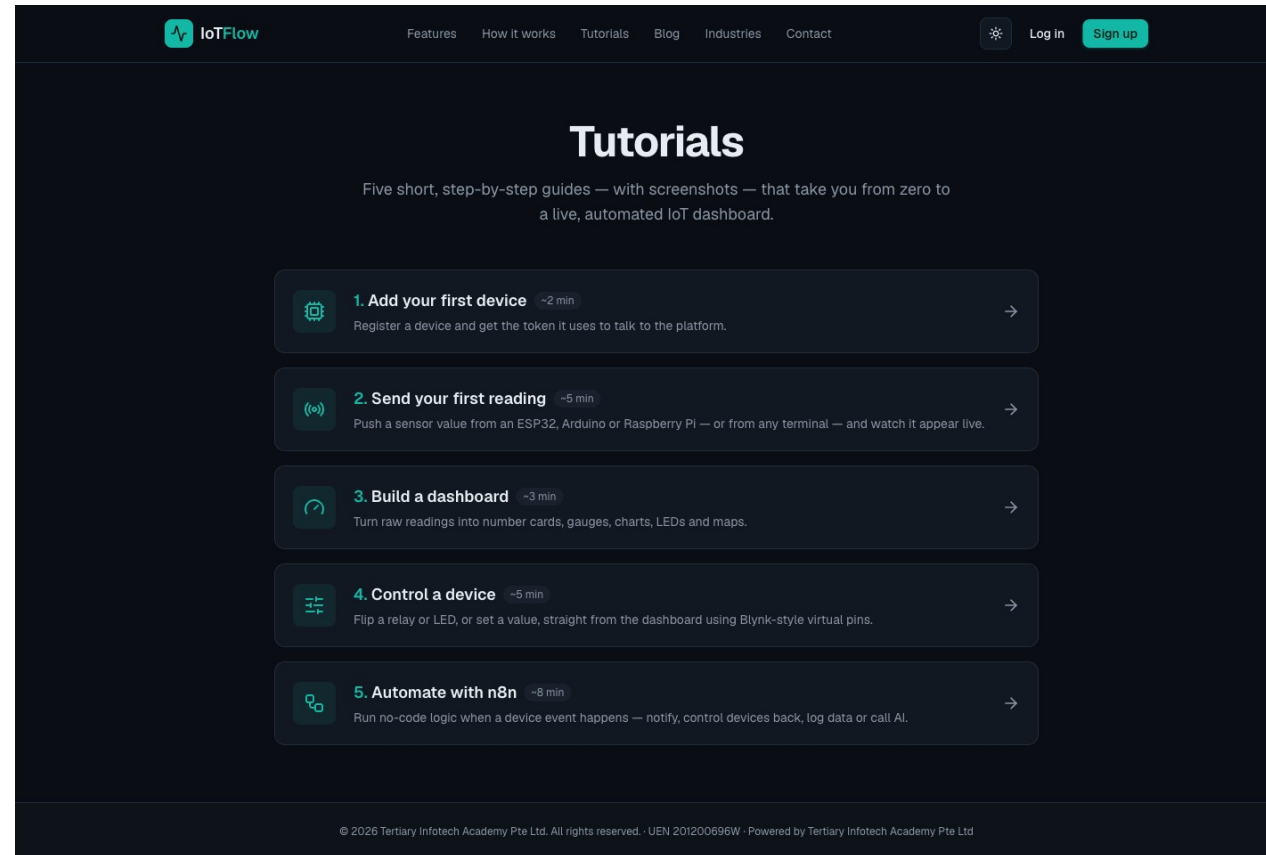
The learner reads device data back out of the cloud — the latest state over the REST API, real-time messages by subscribing to the device's MQTT topics, and per-device telemetry history on the platform — integrating data from multiple sources.

You'll build

A terminal and Python session reading live device state and streaming MQTT messages

Uses: REST API (GET /api/device/state), MQTT subscribe (devices/<id>/down), telemetry history

Read Device Data from the Cloud (REST API & MQTT)



Follow along on the platform: <https://iot.tertiaryinfotech.com/tutorials>

Read Device Data from the Cloud (REST API & MQTT)

STEP 1 OF 6

1

Read your device's latest state from any terminal over the REST API.

```
curl https://iot.tertiaryinfotech.com/api/device/state \  
-H "Authorization: Bearer dev_XXXXXXXXXXXX"
```

Read Device Data from the Cloud (REST API & MQTT)

STEP 2 OF 6



2

Inspect the JSON reply — the platform returns the current value of every metric and virtual pin.

Read Device Data from the Cloud (REST API & MQTT)

STEP 3 OF 6

3

Subscribe to your device's MQTT downlink topic to watch commands and data arrive in real time.

```
mosquitto_sub -h iot.tertiaryinfotech.com -p 1883 \  
-u device -P dev_XXXXXXXXXXXX -t "devices/<id>/down" -v
```

Read Device Data from the Cloud (REST API & MQTT)

STEP 4 OF 6

4

Send a fresh reading (Lab 2) and watch it flow through — uplink telemetry vs downlink commands.

Read Device Data from the Cloud (REST API & MQTT)

STEP 5 OF 6

5

Open the device page on IoTFlow and review its per-device telemetry history and active alerts.

Read Device Data from the Cloud (REST API & MQTT)

STEP 6 OF 6

6

Read the same state from Python and print the values — this is how any external app integrates IoT data.

```
r = requests.get(f"{host}/api/device/state",  
    headers={"Authorization": f"Bearer {token}"})  
print(r.json())
```


Read Device Data from the Cloud (REST API & MQTT)

Test it

The REST call returns the same values shown on the dashboard, and the MQTT subscription prints messages the moment they are published.

Remote Control a Device with Dashboard Virtual Pins

LAB 4

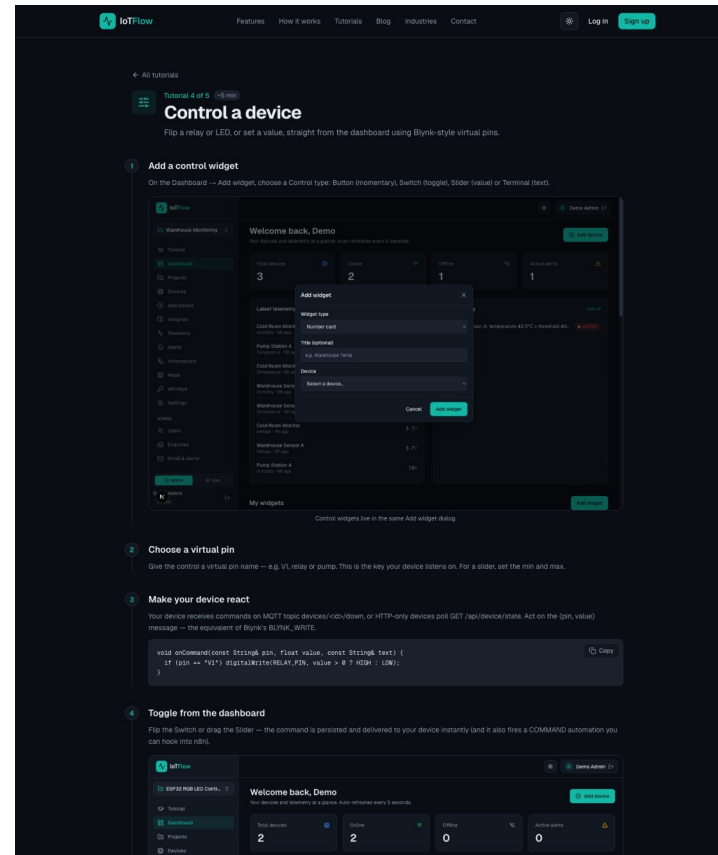
The learner adds Control widgets (button, switch, slider, terminal) bound to virtual pins, implements the device-side command handler, and flips a relay/LED live from the dashboard — Blynk-style two-way control.

You'll build

A dashboard switch and slider that control a relay/LED on the device in real time

Uses: Control widgets (Button, Switch, Slider, Terminal), virtual pins, Python client `@on_command`, MQTT downlink, HTTP polling

Remote Control a Device with Dashboard Virtual Pins



Follow along on the platform: <https://iot.tertiaryinfotech.com/tutorials/control-a-device>

Remote Control a Device with Dashboard Virtual Pins

STEP 1 OF 7



On the Dashboard click Add widget and choose a Control type: Button (momentary), Switch (toggle), Slider (value) or Terminal (text).

Remote Control a Device with Dashboard Virtual Pins

STEP 2 OF 7

2

Assign the virtual pin the widget writes to — e.g. V1, relay or pump. For a slider also set the min/max range.

Virtual pin: V1

Remote Control a Device with Dashboard Virtual Pins

STEP 3 OF 7

3

Understand the downlink path — devices receive commands via MQTT on their command topic, or by HTTP polling.

MQTT topic: `devices/<id>/down` HTTP: `GET /api/device/state`

Remote Control a Device with Dashboard Virtual Pins

STEP 4 OF 7

4

React to commands in Python with the iotflow client — decorate a handler with `@iot.on_command` and poll over HTTP (no extra dependencies).

```
from iotflow import IoTFlow

iot = IoTFlow("https://iot.tertiaryinfotech.com", "dev_XXXXXXXXXXXX", "living-room-sensor")

@iot.on_command
def handle(pin, value, text):
    if pin == "V1":
        print("relay ->", value)    # drive a GPIO here

iot.run(interval=3)                # blocks, polls every 3 s
```

Remote Control a Device with Dashboard Virtual Pins

STEP 5 OF 7

5

For real-time control, use the same handler over MQTT — `loop_forever()` keeps the connection open so commands arrive instantly.

```
iot = IoTFlow(token="dev_XXXXXXXXXXXX", device_id="living-room-sensor",  
              mqtt_host="iot.tertiaryinfotech.com", mqtt_port=1883)
```

```
@iot.on_command  
def handle(pin, value, text): ...
```

```
iot.loop_forever()
```


Remote Control a Device with Dashboard Virtual Pins

STEP 6 OF 7

6

Hardware alternative: implement the same handler in the Arduino sketch and map pins to GPIO outputs.

```
void onCommand(const String& pin, float value, const String& text) {  
  if (pin == "V1") digitalWrite(RELAY_PIN, value > 0 ? HIGH : LOW);  
}
```

Remote Control a Device with Dashboard Virtual Pins

STEP 7 OF 7



Flip the switch and move the slider on the dashboard — commands are delivered to the device immediately; the relay/LED responds. The same events can also trigger automations (Lab 6).

Remote Control a Device with Dashboard Virtual Pins

Test it

Toggling the dashboard switch changes the relay/LED state within a second, and the slider value arrives in your `@on_command` handler as you drag it.

Recap — Read Data and Remote Control from Cloud

- You read device state over REST and streamed live messages with an MQTT subscription.
- You bound Control widgets to virtual pins and handled commands with the Python client's `@on_command` (and Arduino `onCommand`).
- You flipped a relay/LED live from the dashboard — Blynk-style two-way control.
- Alert rules on thresholds or device-offline turn readings into triggers.

Lunch Break

1 hour

TOPIC 04

IoT Data Analytics and Visualization

Dashboards & Widgets · n8n Workflow Automation · AI Analytics · IoT Cybersecurity

Key Concepts — IoT Data Analytics and Visualization

1

Dashboards turn raw readings into number cards, gauges, line/bar charts, LED indicators and maps that auto-refresh.

2

n8n is a low-code workflow automation tool: a device event triggers a flow of nodes that notify, log, call AI or control devices back.

3

AI nodes in a workflow summarise readings, detect anomalies and generate recommendations — analytics with zero code.

4

IoT security matters because the cyber and physical worlds converge: spoofing, tampering, eavesdropping, DoS and elevation of privilege.

Dashboard Widgets

1

Number card

The latest value of a metric, at a glance.

2

Gauge

A value against min/max bounds — great for %, temperature, tank level.

3

Line / Bar chart

Trends over time — the telemetry history plotted live.

4

LED indicator

Binary state — on/off, open/closed, healthy/alarm.

5

Map

Devices plotted by location — fleets, sites, sensors in the field.

6

Device status & alerts

Online/offline state and the list of active alerts.

From Data to Business Insight

- Raw readings become insight when you aggregate, trend and compare them — that is data analytics (K5).
- Dashboards answer 'what is happening now'; history charts answer 'what changed and when'.
- Automations answer 'what should happen next' — thresholds, schedules and AI recommendations.
- Review the data to produce insights of business value (A4): energy saved, downtime avoided, yield improved.

What is n8n?

1

Low-code automation

Drag-and-drop workflow editor — connect 400+ apps with zero code.

2

Nodes

Each node does one job: trigger, transform, notify, store, call AI.

3

Triggers & actions

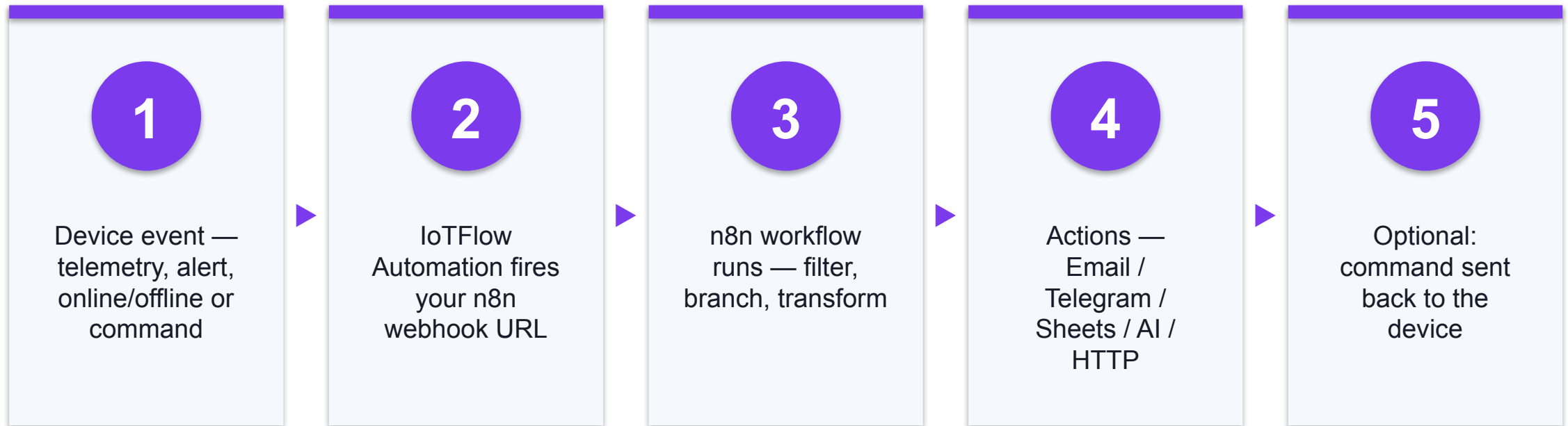
A Webhook trigger starts the flow; action nodes do the work.

4

Workflows

Nodes wired together = a workflow. Activate it and it runs on every event.

IoTFlow + n8n — Event-Driven Automation



AI in Your IoT Workflows

1

Summarise

An AI node turns a day of readings into a one-paragraph brief.

2

Detect anomalies

Ask the model whether the pattern looks abnormal before alerting.

3

Recommend

Generate an action: 'humidity rising — run the dehumidifier 2 h'.

4

Converse

Chat interfaces over your device data — ask questions in plain English.

IoT Security — Why It Matters



Security for IoT adds complexity: the cyber and physical worlds converge.

Threats to IoT Deployments

1

Spoofing

An attacker manipulates device state anonymously or impersonates the originator (MitM).

2

Tampering

Physical attacks — battery drain, sleep deprivation, RNG attacks.

3

Eavesdropping

Intercepting broadcasts to obtain information without authorisation.

4

Denial of Service

Jamming radio or cutting wires renders devices unable to communicate.

5

Elevation of privilege

A device is forced to do something else — a half-open valve tricked fully open.

6

Data breach

Weak credentials and unpatched firmware expose the whole network.

Securing IoT Devices

- Employ device discovery for complete visibility of what is connected.
- Apply network segmentation for a stronger defense.
- Adopt secure password practices — never keep default credentials; protect device tokens.
- Patch and update firmware whenever available.
- Actively monitor IoT devices at all times — alerts and security analytics.

Hands-On Labs — Topic 04

5

Lab 5 — Build a Real-Time IoT Dashboard

A monitoring dashboard with a number card, gauge, chart, LED indicator and map

6

Lab 6 — Automate with n8n — Triggers, Workflows and AI

An n8n workflow fired by device events: threshold alert → notify + log + AI summary

Build a Real-Time IoT Dashboard

LAB 5

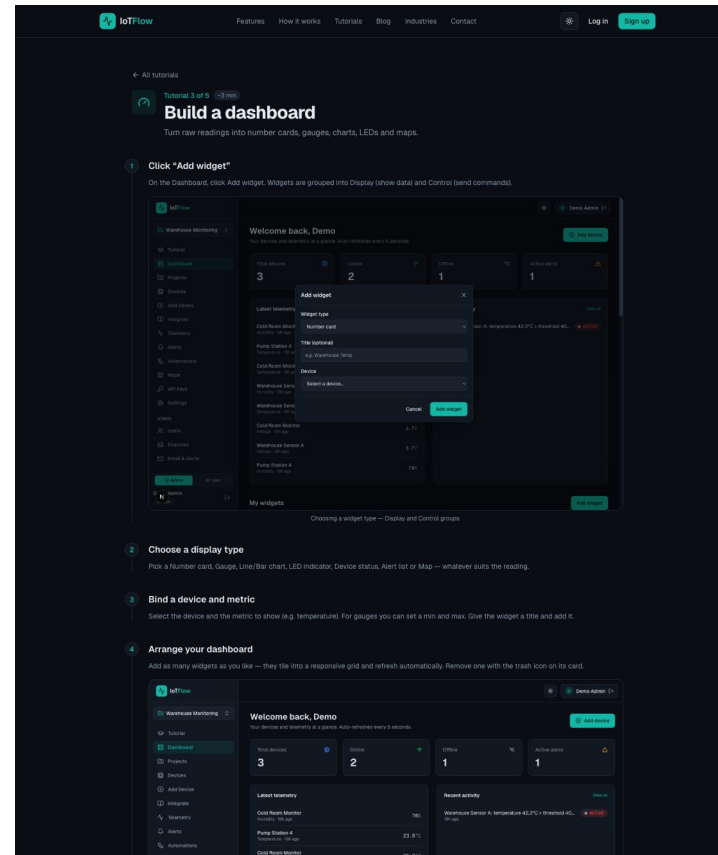
The learner turns raw readings into a live dashboard — number cards, gauges, line/bar charts, LED indicators, device status and maps — arranged on a responsive grid that auto-refreshes.

You'll build

A monitoring dashboard with a number card, gauge, chart, LED indicator and map

Uses: Display widgets: number card, gauge, line/bar chart, LED, device status, alert list, map

Build a Real-Time IoT Dashboard



Follow along on the platform: <https://iot.tertiaryinfotech.com/tutorials/build-a-dashboard>

Build a Real-Time IoT Dashboard

STEP 1 OF 6



On the Dashboard click Add widget — widgets are grouped into Display (show data) and Control (send commands).

Build a Real-Time IoT Dashboard

STEP 2 OF 6

A green circle with a white number 2 inside, indicating the second step in the process.

Add a Number card: select your device, choose the metric (temperature) and give the widget a title.

Build a Real-Time IoT Dashboard

STEP 3 OF 6

3

Add a Gauge for humidity and set sensible min/max bounds (e.g. 0–100 %).

Build a Real-Time IoT Dashboard

STEP 4 OF 6



4

Add a Line chart to plot the temperature history over time.

Build a Real-Time IoT Dashboard

STEP 5 OF 6

5

Add an LED indicator (on/off state) and a Map widget if your device reports a location.

Build a Real-Time IoT Dashboard

STEP 6 OF 6

6

Arrange the layout — widgets snap into a responsive grid, refresh automatically, and can be removed with the trash icon.

Build a Real-Time IoT Dashboard

Test it

Send fresh readings (Lab 2) and watch every widget — card, gauge and chart — update live within the 5-second auto-refresh.

Automate with n8n — Triggers, Workflows and AI

LAB 6

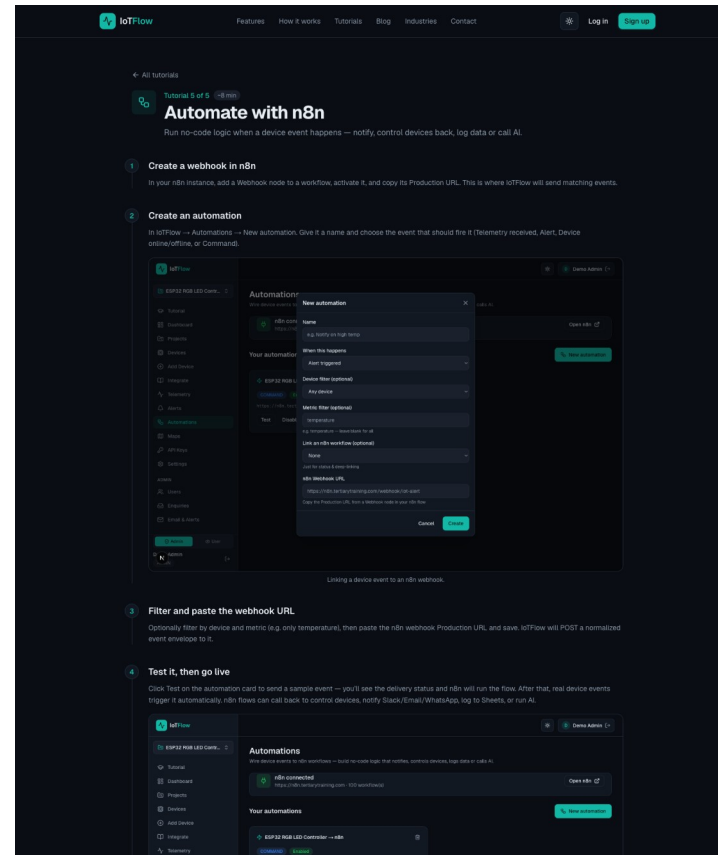
The learner connects IoTFlow to n8n with a webhook, picks a trigger (telemetry, alert, online/offline, command), then builds a no-code workflow that notifies, logs to a sheet, calls AI for insight and even controls the device back.

You'll build

An n8n workflow fired by device events: threshold alert → notify + log + AI summary

Uses: IoTFlow Automations, n8n Webhook node, alert rules, Email/Telegram, Google Sheets, AI nodes

Automate with n8n — Triggers, Workflows and AI



Follow along on the platform: <https://iot.tertiaryinfotech.com/tutorials/automate-with-n8n>

Automate with n8n — Triggers, Workflows and AI

STEP 1 OF 8



In n8n, add a Webhook node to a new workflow, activate the workflow, and copy the webhook's Production URL.

Automate with n8n — Triggers, Workflows and AI

STEP 2 OF 8

2

In IoTFlow, create a new Automation and choose the trigger type: telemetry received, alert fired, device online/offline, or command sent.

Automate with n8n — Triggers, Workflows and AI

STEP 3 OF 8



3

Optionally narrow the trigger by device and metric (e.g. only temperature), then paste the n8n webhook URL and save.

Automate with n8n — Triggers, Workflows and AI

STEP 4 OF 8

4

Create an alert rule on a threshold (e.g. temperature > 30) or device-offline, and hand it off to the automation.

Alert rule: temperature > 30 → fire automation (n8n webhook)

Automate with n8n — Triggers, Workflows and AI

STEP 5 OF 8



5

Use Test to send a sample payload to n8n, inspect the JSON the webhook receives, then enable live triggering.

Automate with n8n — Triggers, Workflows and AI

STEP 6 OF 8



6

In n8n, branch the flow: send an Email/Telegram/WhatsApp notification and append the reading to Google Sheets.

Automate with n8n — Triggers, Workflows and AI

STEP 7 OF 8



Add an AI node to summarise the readings and recommend an action — AI-powered analytics with zero code.

Automate with n8n — Triggers, Workflows and AI

STEP 8 OF 8

8

Close the loop: add an HTTP Request node that sends a downlink command back to the device (e.g. switch a fan on).

```
POST {host}/api/device/command { "pin": "V1", "value": 1 }
```

Automate with n8n — Triggers, Workflows and AI

Test it

Pushing a reading above the threshold fires the n8n flow: the notification arrives, a row is appended to the sheet, the AI summary is generated, and the device reacts to the command sent back.

Recap — IoT Data Analytics and Visualization

- You built a live dashboard — cards, gauges, charts, LEDs and maps on a responsive grid.
- You wired IoTFlow events to an n8n webhook and built a no-code workflow.
- Your flow notified, logged to a sheet, called AI for insight and commanded the device back.
- You know the IoT threat landscape and the practices that secure a deployment.



WRAP-UP

Course Summary & Next Steps

What You Achieved

1

LO1 · IoT technologies

Explained IoT devices, sensors, actuators, triggers and wireless technologies.

2

LO2 · Post data to cloud

Registered a device and streamed telemetry over HTTP and MQTT.

3

LO3 · Control from cloud

Read data via REST/MQTT and controlled devices with virtual pins.

4

LO4 · Analytics & viz

Built dashboards and automated insight with n8n workflows and AI.

Keep Building

- Your IoTFlow account and projects remain yours — <https://iot.tertiaryinfotech.com>.
- Revisit the step-by-step tutorials at <https://iot.tertiaryinfotech.com/tutorials>.
- Connect a real ESP32 or Raspberry Pi at home using the Integrate page snippets.
- Extend your n8n flow: add WhatsApp alerts, Google Sheets logging or an AI agent.

Recommended Courses

- WSQ - Creating High-Converting Email Campaigns with Mailchimp
- WSQ - Data Mining and Machine Learning Fundamentals for Beginners
- WSQ - Business Innovation with Blockchain Technology
- WSQ - Predictive Analytics with PyTorch: Transform Your Data to Prediction
- WSQ - Search Engine Optimization (SEO) for Small and Medium Enterprises

Support

- If you have any enquiries during and after the class, you can contact us:
- Email: enquiry@tertiaryinfotech.com
- Tel: +65 6100 0613
- Course page: <https://www.tertiarycourses.com.sg/wsq-internet-of-things-iot-fundamental-for-beginners.html>

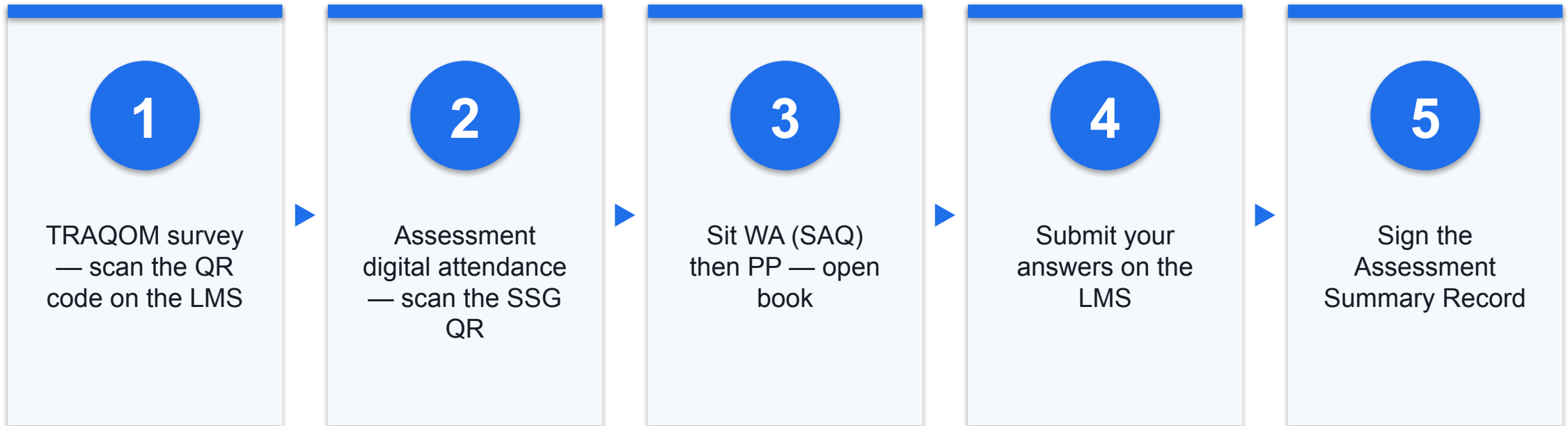
Cert & TRAQOM Survey (Mandatory)

- Complete the TRAQOM survey — scan the QR code on the LMS.
- Your e-certificate is issued via the LMS/TMS after you are assessed Competent.
- Portal: <https://ai-lms-tms.tertiaryinfo.tech/>

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — hands-on IoT platform tasks, 1 hour, open book.
- Format: Open Book — slides, Learner Guide and approved materials only.
- Remember to take the Assessment digital attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Assessment Flow



Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer/administrator displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

SEE YOU IN THE CLOUD

Thank You!

You are now ready to connect, visualise and automate the Internet of Things.